Received: 25 September 2024 | Accepted: 25 September 2024
DOI: 10.1097/HEP.0000000000001112

Practice Guidance

Resmetirom therapy for metabolic dysfunction-associated steatotic liver disease: October 2024 updates to AASLD Practice Guidance

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BACKGROUND

Resmetirom received accelerated approval from the United States Food and Drug Administration (FDA) in March 2024 for the treatment of metabolic dysfunction-associated steatohepatitis (MASH; formerly known as nonalcoholic steatohepatitis or NASH) with moderate to advanced liver fibrosis consistent with stages F2 to F3 fibrosis (F2-F3).[1] The approval was based on findings from the MAESTRO-NASH clinical trial[2] combined with safety data from the MAESTRO-NAFLD-1 trial.[3] In the MAESTRO-NASH trial, treatment with resmetirom achieved the 2 primary endpoints of resolution of steatohepatitis without worsening of fibrosis (26%-30% compared with 10% for resmetirom and placebo, respectively) and improvement in fibrosis without worsening of steatohepatitis at 52 weeks (24%-26% compared with 14% for resmetirom and placebo, respectively).[2]

This article serves as an update to the 2023 AASLD practice guidance on the clinical assessment and management of nonalcoholic fatty liver disease.[4] Since the FDA-approved resmetirom prescribing information[1] does not include implementable guidance for practicing clinicians about whom to treat and how to monitor for response and safety, the writing group (nominated by the AASLD governing board) assimilated data from AASLD practice guidelines on noninvasive liver disease assessments (NILDA) of fibrosis and steatosis,[5,6] published data regarding resmetirom,[2,3,7] recently published expert opinion,[7a] publicly available FDA documents,[8] and selected unpublished data (Madrigal Pharmaceuticals, unpublished data, 2024). While the resmetirom FDA prescribing information represents the official regulatory document to be followed by practitioners, the guidance statements herein were developed to provide timely, implementable recommendations for practicing clinicians. Additional clinical trial and real-world data may necessitate revision of these recommendations in the future.

WHOM TO TREAT

The MAESTRO-NASH trial included participants with metabolic syndrome ($\geq$3 of 5 metabolic risk factors)[9] and stage 1b, 2, or 3 liver fibrosis, and biopsy-confirmed MASH. The resmetirom FDA approval, however, includes only persons with MASH and F2-F3.[1] Notably, the FDA-approved label does not require liver biopsy to

Abbreviations: AASLD, American Association for the Study of Liver Diseases; ALT, alanine aminotransferase; AST, aspartate aminotransferase; F2-F3, stage 2 to 3 liver fibrosis; FDA, United States Food and Drug Administration; GIP, glucose-dependent insulinotropic polypeptide; GLP-1, glucagon-like peptide 1; INR, international normalized ratio; LDL-C, low-density lipoprotein cholesterol; LSM, liver stiffness measurement; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; MRE, magnetic resonance elastography; MRI-PDFF, magnetic resonance imaging proton density fat fraction; NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; NILDA, noninvasive liver disease assessment; T_3, triiodothyronine; T_4, thyroxine; TSH, thyrotropin (aka, thyroid stimulating hormone); VCTE, vibration-controlled transient elastography.

Keywords: fatty liver disease, liver steatosis treatment, steatohepatitis, resmetirom, MASH, MASLD, NAFLD, NASH

confirm the diagnosis of fibrotic MASH. While MASH can only be definitively diagnosed by histologic examination, patient selection in practice is based on evidence of steatosis and fibrosis as determined by NILDA methodologies among persons with cardiometabolic risk factors without other causes of steatosis, notably alcohol consumption of more than 20 g/d for women and more than 30 g/d for men.[10]

There are no FDA-approved noninvasive tests to diagnose MASH with F2-F3 or to monitor response to pharmacotherapy. The aforementioned AASLD guidelines for the use of blood-based and image-based NILDA tests[5,6] in the assessment of liver steatosis and fibrosis were developed in recognition of the limitations of liver biopsy in practice settings. In general, imaging-based NILDAs, such as liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) or magnetic resonance elastography (MRE),[4–6,11–13] yield better accuracy in gauging fibrosis compared with blood-based NILDAs. Magnetic resonance examination utilizing the Liver*MultiScan* image acquisition protocol with corrected T1 imaging has recently been shown to detect fibrotic MASH.[14] However, its utility to confidently rule out clinically inapparent stage 4 fibrosis has not been well demonstrated, thereby limiting its usefulness in selecting candidates for resmetirom therapy.[14]

A detailed discussion of NILDA methodologies to diagnose and quantify hepatic steatosis is beyond the scope of this update. In general, magnetic resonance spectroscopy and magnetic resonance imaging proton density fat fraction (MRI-PDFF) are considered the most accurate quantitative measures of hepatic steatosis, followed by VCTE-controlled attenuation parameter score and gray scale ultrasonography. However, for the purpose of selecting resmetirom treatment candidates, nonquantitative imaging evidence of hepatic steatosis (eg, ultrasonographic evidence) in persons with at least 1 cardiometabolic risk factor and F2-F3 may be sufficient.

In settings where imaging-based NILDA is unavailable, blood-based NILDA may be utilized. Figure 1 describes the use of imaging-based or blood-based NILDA for the selection of patients for whom resmetirom therapy is suitable. As NILDA research evolves, the modalities and thresholds proposed herein to identify MASH with F2-F3 may change.

In the MAESTRO-NASH trial, participants were screened with VCTE and those with LSM >8.5 kPa in whom liver histology demonstrated MASH with F2-F3 were enrolled. Among enrolled participants, the median (interquartile range) LSM by VCTE was 12 kPa (10-15 kPa) and the mean LSM by MRE was 3.5 kPa. The median (interquartile range) of the enhanced liver fibrosis score was 9.7 (9.2-10.4).[2]

These data indicate that there may be individuals who have MASH with F2-F3 who do not fit the criteria for recommended resmetirom therapy (see the green box in Figure 1). For example, 50% of participants with biopsy-proven F2-F3 in the MAESTRO-NASH trial had LSM by VCTE that was outside the interquartile range of 10 kPa to 15 kPa.[2] Thus, practitioners may consider expanded noninvasive criteria in making treatment decisions (see the yellow box in Figure 1). Ultimately, it is the prescriber's prerogative and responsibility to adjudicate a patient's fibrosis status. Utilizing a NILDA test for patient selection at baseline provides the added benefit of allowing for later retesting using the same assay to gauge therapeutic response during or after resmetirom treatment.

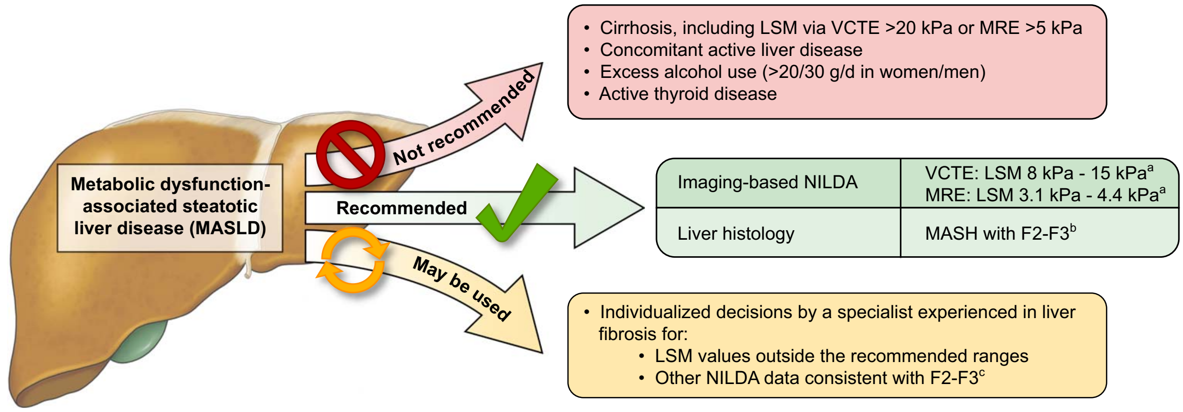
While liver biopsy is not typically recommended for fibrosis staging in current clinical practice, histologic examination remains the gold standard to quantify fibrosis if performed previously (historical biopsy obtained reasonably recently, eg, within 3 years) or for other indications, including suspicion of concomitant liver disease (eg, autoimmune hepatitis). Since NILDA is more readily available than liver biopsy, more current data (eg, within 6-12 months) should be utilized to determine resmetirom treatment candidacy.

CONCOMITANT THERAPY

Regardless of resmetirom treatment status, best practices in metabolic dysfunction-associated steatotic liver disease (MASLD) management should include comprehensive lifestyle modification (nutrition, exercise, and behavior modification) and optimal control of comorbid metabolic conditions.[4,15,16] AASLD guidance previously recognized that in the absence of approved therapy, pharmacologic treatment with semaglutide, pioglitazone, and vitamin E could be considered in appropriate patients.[4]

Glucagon-like peptide 1 (GLP-1) and GLP-1/glucose-dependent insulinotropic polypeptide (GIP) receptor agonists are FDA approved for the treatment of type 2 diabetes and overweight/obesity. They reduce the risk of cardiorenal complications in addition to their effects on glycemic control and weight loss.[17–25] While these pharmaceutical agents are not currently approved for the treatment of MASH, phase 2 randomized, placebo-controlled clinical trials of liraglutide, semaglutide, and tirzepatide have demonstrated their efficacy in reducing steatohepatitis without worsening fibrosis and, in the case of tirzepatide, decreasing fibrosis without worsening of steatohepatitis as well.[26–28] Although not FDA approved for the treatment of MASH, vitamin E and pioglitazone may also improve steatohepatitis.[29–31]

The MAESTRO-NASH trial notably excluded individuals who initiated or dose-modified drugs with GLP-1 receptor agonist activity, thiazolidinediones, or vitamin E within 6 months of randomization.[2] Among persons already taking any of these medications with potential efficacy for treating MASH, data are insufficient to guide



[a] Modified from the AASLD NILDA guidelines.[5]
[b] Liver biopsy is not routinely recommended for staging of MASH.
[c] Imaging-based NILDA is preferred, eg, shear wave elastography (applying local standards for F2-F3) versus enhanced liver fibrosis score (9.2-10.4). The latter range is based on the interquartile range from the MAESTRO trial data; no recommendations are available from the AASLD NILDA guidelines.[6]

FIGURE 1 Selection of Patients for Resmetirom Therapy. Abbreviations: F2-F3, stage 2 to 3 liver fibrosis; LSM, liver stiffness measurement; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease; MRE, magnetic resonance elastography; NILDA, noninvasive liver disease assessment; VCTE, vibration-controlled transient elastography.

prescribers about whether or when to initiate resmetirom therapy. Decisions may be made based on the patient's comorbidity profile, response (if any) to the ongoing therapy, and the fibrosis stage and concern for further liver disease progression. Similarly, in the absence of data supporting specific guidance, among persons receiving neither a GLP-1 or GLP-1/GIP receptor agonist nor resmetirom but who meet treatment criteria for both, clinical judgement may be exercised weighing between concerns for progressive liver fibrosis and the severity of extrahepatic comorbidity. This is not to be construed as an endorsement of off-label therapy for MASH but rather an acknowledgement of uncertainty about comparative efficacy of these drugs in light of the MAESTRO-NASH exclusion criteria.

Resmetirom is a substrate for cytochrome P450 enzyme 2C8 and organic anion transporting polypeptides 1B1 and 1B3, and a weak cytochrome P450 enzyme 2C8 inhibitor.[1] If resmetirom is used concurrently with a cytochrome P450 2C8 inhibitor (a common example being clopidogrel), a resmetirom dose reduction is recommended (80 mg/d for persons who weigh 100 kg or more; 60 mg/d for persons who weigh less than 100 kg). Given the comorbidity profile of persons with MASLD, cardiovascular risk management is an important aspect of medical management, including the use of statin pharmacotherapy. Because resmetirom is a substrate for organic anion transporting polypeptides 1B1 and 1B3, concomitant administration of resmetirom and a statin may influence statin metabolism. When taken concurrently with resmetirom, the recommended maximum dosage for rosuvastatin and simvastatin is 20 mg/d; the recommended maximum dosage for atorvastatin and pravastatin is 40 mg/d. Notably, resmetirom decreases low-density lipoprotein but its long-term effects on cardiovascular disease risk remain to be determined.[2,3,21]

ADVERSE EVENTS AND SAFETY MONITORING

Table 1 and Figure 2 summarize recommendations for safety and efficacy monitoring of persons receiving resmetirom. In both the MAESTRO-NASH and the MAESTRO-NAFLD-1 trials, the most common treatment-emergent adverse events were diarrhea and nausea, which developed in 24% to 34% and 12% to 22% of resmetirom-treated participants, respectively.[2,3] Typical onset was less than 12 weeks after initiation of resmetirom therapy and often less than 4 weeks. While median duration of diarrhea was 15 days to 20 days and there were no cases of severe diarrhea, the symptoms lasted greater than 4 weeks in approximately half of the participants who developed diarrhea. Resmetirom was associated with increased incidence of symptomatic gallstone disease, although the presence of gallstones and/or a history of biliary colic are not contraindications to resmetirom treatment.

There was a sole case of severe hepatotoxicity in a participant receiving resmetirom. While on therapy, the patient developed jaundice (bilirubin 15 times the upper limit of normal)[1] and biopsy showed interface hepatitis consistent with autoimmune hepatitis versus drug-induced autoimmune-like hepatitis. In retrospect, this patient met the criteria for primary biliary cholangitis and had strongly positive antimitochondrial and antinuclear antibodies at baseline. The adjudication committee assessed a 25% to 50% likelihood that resmetirom

TABLE 1 Safety and efficacy assessments at baseline and during 12 months of treatment with resmetirom

	Safety/Efficacy assessments	Safety assessments		Efficacy assessments	
Timeframe	**Hepatic function panel[a]**	**Thyroid function[b]**	**Lipid profile[c]**	**Noninvasive measurement of liver stiffness[d]**	**MRI-PDFF[e]**
Before treatment initiation	✓	✓	✓	✓	Consider
3 months	✓				
6 months	✓	✓	✓		
12 months	✓	✓	✓	Repeat if imaging NILDA was used at baseline	Consider repeating if baseline data are available

Color Key Green: Recommended for all treated patients.
Blue: Recommended for a subset of patients for whom testing is appropriate.
Yellow: Optional assessments based on availability.
[a]Hepatic function panel (total protein, albumin, alkaline phosphatase, bilirubin (total and direct), ALT, and AST) should be monitored for safety. See Figure 2 for use of ALT as a response criterion.
[b]Resmetirom was associated with mild reductions in TSH and free T_4, particularly in persons receiving levothyroxine replacement therapy. Monitoring thyroid function with TSH and free T_4 testing is recommended for patients receiving thyroid hormone replacement therapy.
[c]Resmetirom treatment resulted in lower LDL-C levels than treatment with placebo. Titration of statins may be necessary, with or without additional lipid-lowering treatments, depending on individual cardiovascular risk and LDL-C targets.
[d]If available, image-based measurement of liver stiffness is strongly preferred at baseline. Repeat the same modality at 12 months.
[e]Less than 30% improvement (ie, reduction) in MRI-PDFF from baseline at 12 months was strongly associated with lack of histologic response in the MAESTRO-NASH trial. Changes in controlled attenuation parameter as measured by VCTE are not associated with histologic treatment response and should not be used to assess futility of resmetirom treatment.
Abbreviations: MRI-PDFF, magnetic resonance proton density fat fraction; NILDA, noninvasive liver disease assessment.

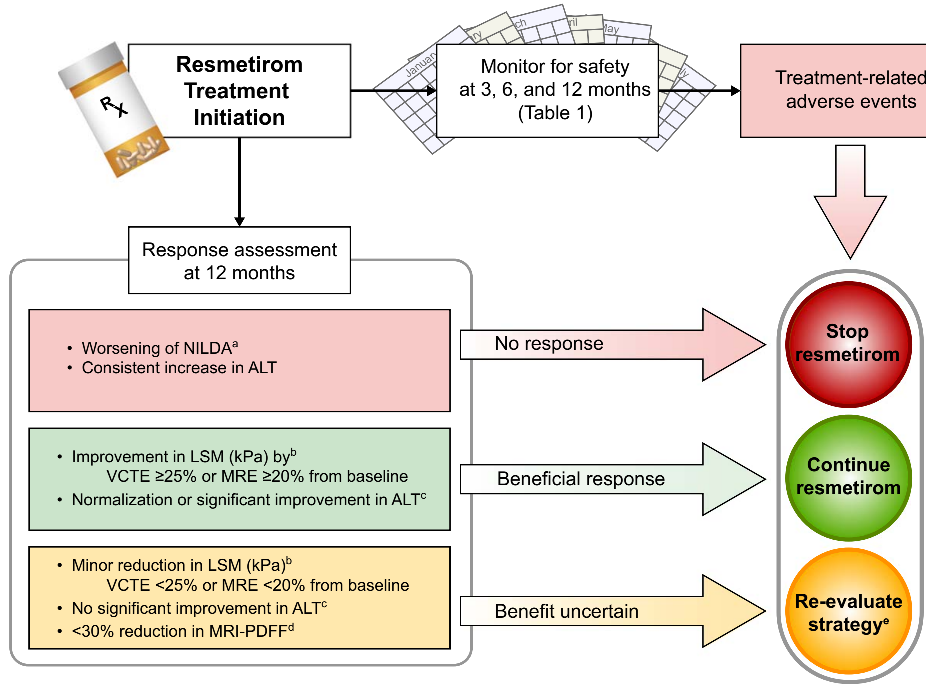
contributed to this hepatoxicity (Madrigal Pharmaceuticals, unpublished data, 2024).

Regular hepatic function panel testing during resmetirom therapy is recommended to screen for hepatotoxicity. Resmetirom should be discontinued if clinically-significant hepatotoxicity develops as defined per the AASLD drug-induced liver injury practice guidance, specifically: AST or ALT greater than 5 times the upper limit of normal or alkaline phosphatase greater than 2 times the upper limit of normal (or pretreatment baseline if baseline is abnormal) on 2 occasions; or total serum bilirubin greater than 2.5 mg/dL plus elevated AST, ALT, or alkaline phosphatase; or INR greater than 1.5 plus elevated AST, ALT, or alkaline phosphatase.[32]

ENDOCRINOLOGIC CONSIDERATIONS

Hypothyroidism occurs commonly in persons with hepatic steatosis, although its relationship with steatohepatitis has been less consistent and confounded by insulin resistance, obesity, and type 2 diabetes.[33] For example, 13% of participants in the MAESTRO-NASH trial were receiving treatment for hypothyroidism at enrollment with thyroid hormone therapy dosages up to 75 ug/d. Persons with active hyperthyroidism or untreated hypothyroidism (ie, TSH > 10 mIU/L without symptoms, or TSH > 7 mIU/L with symptoms) were excluded from the MAESTRO-NASH study. In the trial, the mean plasma free T_4 concentration decreased approximately 16% to 21% (up to 24% in those receiving levothyroxine), although rarely leading to abnormal levels.[2] TSH and T_3 also changed slightly but consistently remained within the normal ranges.[3] Development of hypothyroidism requiring levothyroxine replacement occurred in 1.8% of participants receiving resmetirom.[2] The FDA integrated review of resmetirom concluded that thyroid axis function was maintained during resmetirom therapy and recommended no thyroid-specific management.[8]

These data indicate that abnormal thyroid function (hyperthyroidism or hypothyroidism) should be addressed before initiating resmetirom therapy, which necessitates thyroid function assessment with at least a TSH level at baseline or up to 6 months prior. In



[a] Assess based on the same imaging-based or blood-based markers used to determine treatment eligibility.
[b] LSM improvement thresholds of VCTE ≥25% or MRE ≥20% are based on assay characteristics and not specifically validated for clinical decisions in resmetirom treatment patients. There are currently no comparable data to determine response in blood-based NILDAs.
[c] Applies to patients with elevated ALT at baseline. No specific ALT response cutoffs are available from the MAESTRO trial.
[d] MRI-PDFF reduction by >30% does not necessarily correlate with histologic response.
[e] Options may include re-optimizing lifestyle interventions and considering other therapy, with or without stopping resmetirom.

FIGURE 2 Assessment for Treatment Outcome in Patients Receiving Resmetirom. Abbreviations: ALT, alanine aminotransferase; LSM, liver stiffness measure; MRE, magnetic resonance elastography; MRI-PDFF, magnetic resonance imaging proton density fat fraction; NILDA, noninvasive liver disease assessment; VCTE, vibration-controlled transient elastography.

accordance with FDA guidance,[8] persons without thyroid function abnormalities at baseline do not require systematic monitoring of thyroid function tests while receiving resmetirom—although clinical vigilance should be exercised to discern and address emergence of adverse effects as with any pharmacologic therapy. Among persons with known thyroid disease, standard clinical monitoring (eg, TSH and free T_4 at 3-month to 6-month intervals) should be followed at the discretion of the treating clinician (see Table 1).

Resmetirom increases sex hormone binding globulin as seen in the MAESTRO-NAFLD-1 and MAESTRO-NASH trials.[2,3] As expected with an increase in sex hormone binding globulin, mean total testosterone in males increased approximately 50% to 100%; free testosterone, however, was unchanged overall. Significant increases of total estradiol were observed in females but data on free estradiol were not reported. The clinical significance of any possible effects of these hormonal changes remains to be established.

Management of dyslipidemia remains an essential element in the holistic care of persons with MASH, especially in the presence of type 2 diabetes and/or known atherosclerotic complications. Among persons in whom a statin dosage is reduced as recommended at the initiation of resmetirom therapy, care must be taken to continue to meet cardiovascular risk management targets by monitoring the serum lipid profile (see Table 1) and adjusting statin therapy accordingly during resmetirom treatment.

IDENTIFYING EFFICACY AND FUTILITY OF RESMETIROM TREATMENT

Resmetirom received accelerated initial FDA approval with final approval contingent upon verification and description of long-term clinical benefit in the ongoing MAESTRO-NASH trial.[8] As of the time of writing this guidance update, these long-term data are not yet available. The 52-week trial data documented MASH resolution and fibrosis improvement in 26% to 30% and 24% to 26% of treated participants, respectively.[2] Among the remaining greater than 70% of study participants, the primary treatment endpoints were not reached as of week 52.[2]

Table 1 and Figure 2 summarize recommendations for safety and efficacy monitoring of persons receiving resmetirom. Data are limited from the MAESTRO-NASH trial regarding changes in specific NILDA tests associated with histologic response. Similarly, reliable evidence to correlate LSM data with histologic changes in liver fibrosis is lacking. In general, the AASLD NILDA practice guideline suggests against use of imaging-based NILDA as a standalone test to assess regression or progression of liver fibrosis.[5] With that caveat in mind, some available data may be used to make tentative recommendations.

According to the Radiological Society of North America Quantitative Imaging Biomarkers Alliance, a 19% reduction in LSM by MRE is considered to reflect a true reduction in liver fibrosis.[34] In the MAESTRO-NASH trial, the secondary endpoint of a 20% or greater reduction in MRE-measured stiffness yielded response rates similar to rates of improvement in histologic fibrosis,[2] suggesting a 20% or greater change in kPa on MRE may be useful to define a meaningful change for improvement and, by extension, for worsening (see Figure 2).

A 20% reduction in VCTE LSM or a decrease in LSM to less than 10 kPa has been proposed as a clinically meaningful decrease that is associated with a lower risk of complications,[35–37] although this has not been evaluated in the context of drug trials. In the MAESTRO-NASH trial, a 25% decrease in VCTE LSM overestimated histologic response.[2] Forty-nine percent of participants treated with resmetirom had at least a 25% decrease in VCTE LSM compared with 26% to 30% and 24% to 26% who had MASH resolution and fibrosis improvement, respectively. Notably, histologic response was also seen in some people with no VCTE LSM improvement.[7] Hence, VCTE LSM is not an ideal marker to monitor response to resmetirom. However, the writing group recognizes that many clinicians and patients might use VCTE LSM to make treatment decisions. In the absence of direct data to support a specific threshold and for the purposes of this guidance, the writing group suggests using a 25% decrease in VCTE LSM to assess for response to resmetirom. By extension, a 25% increase in VCTE LSM may define meaningful worsening. These thresholds may change as new evidence emerges.

Improvement in the enhanced liver fibrosis score and other blood-based NILDA has not correlated consistently with resolution of MASH or regression of fibrosis in clinical trials, whereas worsening of the biomarkers from baseline may be construed as evidence of futility.[38]

Given its limited availability in routine clinical practice, quantitative assessment of steatosis by MRI-PDFF should not be a requirement for initiating or continuing resmetirom therapy. Among persons who have access to this technology at baseline, a repeat test after 52 weeks of therapy may provide useful information to assess the utility of continuing resmetirom therapy. In the MAESTRO-NASH trial, participants treated with resmetirom who did not have a 30% or greater reduction in steatosis on MRI-PDFF by week 52 had a histologic response rate that did not differ from that of the placebo group (approximately 10%).[2] Thus, among persons who do not achieve a 30% or greater reduction in MRI-PDFF after 1 year of resmetirom therapy, clinicians should take into account these data in determining the utility of ongoing therapy and likelihood of success with continued treatment. In contrast, a 30% or greater reduction (improvement) in steatosis as determined by MRI-PDFF among MAESTRO-NASH participants was associated with improved histologic outcomes compared with placebo,[2] similar to findings from other studies.[39,40] Of note, among persons who achieved this response on resmetirom, histologic response rates were still less than 50%, which the writing group considers a positive predictive value insufficiently high to be used alone to predict histologic efficacy.[41]

Changes in the controlled attenuation parameter have not been directly tested as a predictor of histologic response in clinical trials and was not predictive of response in the MAESTRO-NASH trial.[2] Future analyses of data from MAESTRO-NASH and other clinical trials may provide additional information on the value of NILDA for treatment monitoring.

Improvement in ALT was weakly correlated with rates of histologic response in the MAESTRO-NASH trial.[7] In prior studies among persons with elevated baseline ALT levels, ALT changes associated with MASH resolution without worsening of fibrosis included a 17 U/L or greater decrease in ALT,[39] or reduction of ALT to 40 U/L or less and by at least 30% from baseline.[42] Based on these data, normalization or significant improvement in ALT may be considered a favorable response, while resmetirom-treated persons whose ALT remains consistently elevated may be judged as nonresponders.

Figure 2 synthesizes these monitoring data. Using the MRE and VCTE LSM thresholds defined above, persons with fibrosis improvement as well as those with favorable ALT response should continue resmetirom therapy. Among persons with significantly worsening NILDA for fibrosis or ALT level, discontinuation of resmetirom should be considered.

Among persons who do not meet criteria for continuation or discontinuation after 12 months of resmetirom therapy, available data are insufficient to make a firm recommendation regarding whether to continue resmetirom. An important consideration is that stabilization of fibrosis may portend important long-term benefits; for persons with advanced fibrosis, apparent lack of fibrosis improvement should not mandate discontinuation of resmetirom therapy. Thus individualized, shared decision-making with the patient is recommended, holistically incorporating baseline fibrosis status; comorbidity profile associated with progressive liver disease; concomitant therapy including adherence and response to lifestyle interventions; changes in liver biochemical and NILDA parameters; and adverse effects (if any) during resmetirom therapy. The decision may include re-optimizing lifestyle interventions and considering other therapy, with or without continuing resmetirom.

PRACTICE RECOMMENDATIONS

Patient selection

- Resmetirom can be considered for treatment of adults with MASH and moderate to advanced liver fibrosis (consistent with F2-F3). Criteria for treatment eligibility include:
  - Noninvasive liver disease assessment—preferably imaging-based test results—consistent with MASH with F2-F3, or
  - Historical liver biopsy demonstrating MASH with F2-F3 (and without evidence of concomitant, histologically active autoimmune liver disease).
- The recommended dosage of resmetirom is 100 mg/d for persons who weigh 100 kg or more, or 80 mg/d for persons who weigh less than 100 kg.
- If resmetirom is used concurrently with a moderate cytochrome P450 2C8 inhibitor (eg, clopidogrel), the recommended dosage is 80 mg/d for persons who weigh 100 kg or more or 60 mg/d for persons who weigh less than 100 kg.

Pretreatment considerations

- Resmetirom is not recommended for persons with compensated or decompensated cirrhosis, concomitant uncontrolled active liver diseases (such as autoimmune hepatitis and primary biliary cholangitis), or ongoing alcohol consumption greater than 20 g/d for women or greater than 30 g/d for men.
- Thyroid function assessment is recommended before initiating resmetirom treatment. For persons with untreated hyperthyroidism or hypothyroidism, resmetirom initiation is not recommended until thyroid function is optimized.
- Resmetirom initiation is not recommended for patients with symptomatic gallstone-related conditions such as acute cholecystitis.

On-treatment monitoring

- Hepatic function panel testing should be obtained at baseline and at periodic intervals (eg, 3, 6, and 12 months) to determine response and adverse events while on resmetirom therapy. Resmetirom should be discontinued if hepatotoxicity develops, as defined by the AASLD drug-induced liver injury practice guidance.[32] There are insufficient data to make a recommendation on monitoring after 12 months of resmetirom treatment but continued monitoring with hepatic function panel testing every 6 months is suggested.
- Among persons with known thyroid disease, standard laboratory monitoring (eg, TSH and free T_4) per established guidelines is recommended while receiving resmetirom therapy.
- Resmetirom can be used concurrently with statins; however, practitioners should be aware of the maximum recommended daily dosages, including atorvastatin 40 mg/d, pravastatin 40 mg/d, rosuvastatin 20 mg/d, and simvastatin 20 mg/d. Continued attention to comorbidity management including hyperlipidemia is recommended, particularly if the statin dose is modified at the outset of resmetirom therapy.

Determination of efficacy and futility

The following recommendations are based on data at 52 weeks after resmetirom treatment initiation.[2] Since the MAESTRO-NASH trial is ongoing, these recommendations may change when longer-term data become available. The decision to continue treatment beyond 12 months should be based on assessing whether a patient has evidence of improvement, worsening or treatment failure, or disease stabilization.

- In persons whose treatment candidacy was determined by liver stiffness measurements (VCTE or MRE), a repeat measurement at 12 months of therapy is recommended to assess for treatment response. In the absence of resmetirom-specific data, the writing group suggests that improvement (or worsening) of VCTE of at least 25% or MRE of at least 20% from baseline represents a significant (ie, clinically meaningful) change beyond intraindividual measurement variability.
- Continuation of resmetirom treatment in persons with significant fibrosis improvement is recommended.
- Among persons with evidence of worsening liver disease at 12 months of treatment, discontinuation of resmetirom should be considered. These individuals can be identified by clinical data suggestive of worsening liver disease including consistent increase in ALT or fibrosis progression as assessed by NILDA.
- Among persons without clear evidence of liver disease improvement or worsening, the decision to continue therapy (pending longer-term data from ongoing studies) should be based on holistic assessment. Clinicians and patients should consider: the baseline fibrosis status and the potential benefit of slowing or stabilizing liver fibrosis even if no regression is noted; the individual comorbidity profile associated with progressive liver disease;

concomitant therapy including adherence and response to lifestyle interventions; changes in liver biochemical and NILDA parameters; and adverse effects (if any) during resmetirom therapy.

AUTHOR CONTRIBUTIONS

All authors contributed to the conceptualization, intellectual content, development of the original manuscript, and critical revisions. All authors provided approval of the final manuscript for publication.

ACKNOWLEDGMENTS

The authors appreciate the expert review by Drs Elizabeth Verna and Cynthia Levy, the Chair and Vice Chair, respectively, of the Practice Guidelines Committee of the American Association for the Study of Liver Disease. The authors thank Madrigal Pharmaceuticals for providing unpublished data. Madrigal Pharmaceuticals and its staff did not provide funding support and were not involved in the writing or approval of this manuscript.

FUNDING INFORMATION

Funding for the development of this practice guidance update was provided exclusively by the American Association for the Study of Liver Diseases.

CONFLICTS OF INTEREST

Vincent L. Chen was awarded research grants, paid to his institution, from AstraZeneca, Kowa Pharmaceuticals America, and Ipsen Biopharmaceuticals; his spouse is employed by CVS Health. Timothy R. Morgan has no conflict to report. Yaron Rotman's spouse is employed by AstraZeneca. Heather M. Patton has no conflict to report. Kenneth Cusi is a consultant for Aligos Therapeutics, Arrowhead Pharmaceuticals, AstraZeneca, 89Bio Inc, Bristol Myers Squibb, Boehringer Ingelheim, Covance Laboratories Inc, Eli Lilly and Company, Novo Nordisk, ProSciento, Sagimet Biosciences, Siemens USA, and Terns Pharmaceuticals; he was awarded research grants, paid to his institution, from Boehringer Ingelheim, Echosens, Inventiva Pharma, Labcorp, and Perspectum. Fasiha Kanwal was awarded research grants, paid to her institution, from Gilead Sciences Inc, and Merck and Company Inc. W. Ray Kim has no conflict to report.

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How to cite this article: Chen VL, Morgan TR, Rotman Y, Patton HM, Cusi K, Kanwal F, et al. Resmetirom therapy for metabolic dysfunction-associated steatotic liver disease: October 2024 updates to AASLD practice guidance. Hepatology. 2025;81:312–320. https://doi.org/10.1097/HEP.0000000000001112